

 Population vs Sample Bayes' Theorem & ML

INSTANT

 Data Analysis Diploma • Session 15

INFERENCEAL STATISTICS & PROBABILITY THEORY

Population → Sample → Inference → Decision · Probability · Bayes' Theorem ·
CLT · Confidence Intervals · Hypothesis Testing · A/B Testing

7 Parts

Inferential Mastery
Probability to A/B Testing

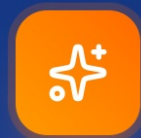
48 Slides

Interactive Presentation
Worked Examples & Exercises

A/B Testing

Production Experiments
Sample Size, Power & Pitfalls

 CLT & Confidence Intervals Hypothesis & A/B Testing



01 • FROM DATA TO DECISIONS

From Data to Decisions

How we draw rigorous conclusions about the world from limited observations

— DESCRIPTIVE VS INFERENCE

Two Branches of Statistics

Understanding where Inferential Statistics fits in the data science pipeline

DESCRIPTIVE STATISTICS

Summarises and describes data you already have.

- You describe what IS in your sample
- Mean, Median, Mode
- Standard Deviation, Variance
- Charts and distributions

Goal: Understand your current data.

Note from Bruce et al. (Practical Statistics): 'Classical statistics focused almost exclusively on inference — a complex set of procedures for drawing conclusions about large populations based on small samples.'

INFERENCE STATISTICS

Uses sample data to make conclusions about a larger population.

- You infer what IS LIKELY true beyond your sample
- Probability Theory & Bayes' Theorem
- Confidence Intervals
- Hypothesis Tests & A/B Testing

Goal: Make decisions about the unknown.

At the heart of statistics lies variability: measuring it, reducing it, distinguishing random from real variability, and making decisions in the presence of it.

— POPULATION (N) VS SAMPLE (n)

Population vs Sample — The Core Concept

Every statistical inference starts with understanding parameters vs statistics

POPULATION (N)

Target Group

The entire group you want to draw conclusions about.

Parameters (True values, often unknown):

μ (mu) = population mean
 σ (sigma) = population std deviation
 p = population proportion

Examples: All customers of e-commerce platform, All adults in a country, All transactions ever made.

SAMPLE (N)

Observed Subset

A subset of the population actually observed.

Statistics (Estimates from sample):

$\bar{x}$ = sample mean
 s = sample std deviation
 $\hat{p}$ = sample proportion

Examples: 1,000 randomly selected customers, 500 adults surveyed, Last month's transactions.

POPULATION (N)

--- Sampling --->>>

SAMPLE (n)

--- Inference --->>>

DECISION / PARAMETER ESTIMATE

— HOW YOU COLLECT MATTERS

Types of Sampling — How You Collect Matters

A biased sample leads to wrong inferences, no matter how good your math is:



[1] Simple Random Sampling

Every member has an equal chance of selection.

Advantages: Gold standard; minimises bias

Limitation: Needs a complete population list

Example: Randomly select 500 customers from full CRM database.



[2] Systematic Sampling

Select every k -th element from the population list.

Advantages: Simple; evenly spread

Limitation: Can miss periodic patterns

Example: Survey every 10th customer who visits the site today.



[3] Stratified Sampling

Divide population into groups (strata), then randomly sample each.

Advantages: Ensures all subgroups are represented

Example: Survey 200 from each age group: 18–30, 31–50, 51+.



[4] Cluster Sampling

Divide into clusters, randomly select whole clusters.

Advantages: Cost-effective for large populations

Limitation: Higher sampling error

Example: Randomly choose 20 cities; survey all residents.



[5] Convenience Sampling

Select whoever is easiest to reach.

Limitation: High risk of bias; results not generalisable

Only acceptable for exploratory, early-stage work
Example: Surveying Twitter followers about a product.



[6] Purposive Sampling

Select based on researcher judgement about who is most useful.

Limitation: Subjective; researcher bias risk

Useful when specific expertise is needed

Example: Interviewing only senior data scientists about MLOps.

— EXERCISE 01 · FOUNDATIONS

Exercise — Foundations

A data analyst wants to study average monthly spending of users on a streaming platform with 4 million registered users worldwide.

Q1: Identify the Population and Parameter of interest.

Hint: What is the full group? What number are you trying to estimate?

Reveal Solution ▾

Q2: The analyst surveys 2,000 randomly selected users. What is the Sample and Statistic?

Hint: What is the subset? What calculated value estimates the parameter?

Reveal Solution ▾

Q3: The analyst only surveys users who logged in last week. Which sampling method is this? What bias does it introduce?

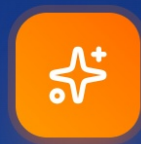
Hint: Think about who is excluded and whether they differ systematically.

Reveal Solution ▾

Q4: Suggest a better sampling method and explain why it produces more reliable inferences.

Hint: Which method ensures all user types (heavy, light, inactive) are represented?

Reveal Solution ▾



02 • QUANTIFYING UNCERTAINTY

Quantifying Uncertainty

The mathematical language for describing how likely events are to occur

— BUILDING THE VOCABULARY

What Is Probability?

Building the core vocabulary before applying the rules: $P(E) = \text{Favourable} / \text{Total}$ ($0 \leq P(E) \leq 1$)

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— ADDITION RULE · $P(A \text{ OR } B)$

Addition Rule — $P(A \text{ or } B)$

Calculating the probability that at least one of two events occurs

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— MULTIPLICATION RULE · $P(A \text{ AND } B)$

Multiplication Rule — $P(A \text{ and } B)$

Probability of two events both occurring: Dependent vs Independent Events

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Probability of two events both occurring: Dependent vs Independent Events

— $P(A | B) = P(A \cap B) / P(B)$

Conditional Probability — $P(A \text{ given } B)$

'Given that B happened, how likely is A?' — Restricting the sample space to B

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— SPAM FILTER CASE STUDY

Applied Example — Email Spam Filter

Scenario: 10,000 emails (2,000 spam, 8,000 legit). 1,800 spam contain "FREE". 400 legit contain "FREE".

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— EXERCISE 02 · PROBABILITY RULES

Exercise — Probability Rules

A survey of 1,000 job applicants shows: 400 have a degree (D), 300 have work experience (E), 150 have both.

Q1: Calculate $P(D \text{ or } E)$ — probability of having a degree OR experience.

Hint: Use Addition Rule: $P(D \cup E) = P(D) + P(E) - P(D \cap E)$.

Reveal Solution ▾

Q2: Are D and E independent events? Show your reasoning with numbers.

Hint: Check: if independent, $P(D \cap E)$ should equal $P(D) \times P(E)$.

Reveal Solution ▾

Q3: Given an applicant has experience, what is the probability they also have a degree?

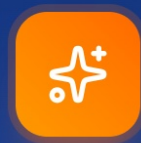
Hint: Use Conditional Probability: $P(D|E) = P(D \cap E) / P(E)$.

Reveal Solution ▾

Q4: If two applicants are selected WITH replacement, what is $P(\text{both have degrees})$?

Hint: With replacement, selections are independent. Use Multiplication Rule for independent events.

Reveal Solution ▾



03 • UPDATING BELIEFS WITH EVIDENCE

Updating Beliefs with Evidence

The most powerful idea in probability — and the foundation of modern AI

— INTUITION BEFORE THE FORMULA

Bayes' Theorem — The Intuition First

Bayes' Theorem answers: How should I update my belief about something after seeing new evidence?

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— $P(A|B) = P(B|A) \times P(A) / P(B)$

Bayes' Theorem — The Formula Built Step by Step

Posterior = (Likelihood x Prior) / Marginal Evidence Normaliser

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Posterior = (Likelihood x Prior) / Marginal Evidence Normaliser

— MEDICAL TEST CASE STUDY

Bayes in Action — Medical Test (Full Worked Example)

Why a 99% accurate test on a rare disease (1% rate) results in only a 50% actual positive probability

INTERACTIVE BAYES SIMULATOR

Disease Prevalence (Prior $P(\text{Sick})$): 1.0%



Test Sensitivity (True Positive Rate): 99.0%



Test Specificity (True Negative Rate): 99.0%



Key Insight: When a disease is rare (e.g. 1%), even a 99% accurate test yields a ~50% false alarm rate because healthy people vastly outnumber sick people!

BAYES CALCULATION OUTPUT

Prior $P(\text{Sick}) = 0.0100$

Likelihood $P(\text{Test+} \mid \text{Sick}) = 0.9900$

False Positive $P(\text{Test+} \mid \text{Healthy}) = 0.0100$

Total Evidence $P(\text{Test+}) = 0.0198$

Posterior $P(\text{Sick} \mid \text{Test+})$:

50.0%

If you test positive, there is a **50.0%** actual probability you have the disease!

— PRODUCTION ALGORITHMS

Bayes' Theorem in Machine Learning

Bayes' theorem is the foundation of an entire family of ML models and Bayesian statistics:



[NB] Naive Bayes Classifier

Direct application of Bayes to classification. Assumes features are conditionally independent.

Used for: Spam filtering, sentiment analysis, document classification.

Fast, interpretable, works with small datasets.



[BI] Bayesian Inference

Gives a probability distribution over parameter values instead of a point estimate.

Uncertainty is explicit and updates continuously as data arrives.

Used in A/B testing, recommendation systems.



[PD] Posterior Distributions

In Bayesian ML, parameters have probability distributions. Posterior = what we believe about a parameter after seeing data.

Used in Bayesian Neural Networks, uncertainty estimation.



[BO] Bayesian Optimisation

Uses a Bayesian model to decide which hyperparameter to try next.

More efficient than grid search.

Standard in AutoML, neural architecture search.



[RC] Recommendation Systems

$P(\text{User likes item} \mid \text{User history})$

Prior: What most users like

Likelihood: How user ratings update belief

Netflix, Spotify, Amazon use variants of this.

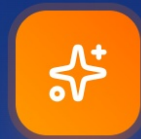


[AB] Bayesian A/B Testing

Asks: "What is $P(B \text{ is better than } A)$?" instead of $p\text{-value} < 0.05$.

More intuitive for business decisions.

Allows stopping test early when certainty is sufficient.



04 • THE MOST IMPORTANT THEOREM

The Most Important Theorem in Statistics

Why the normal distribution appears everywhere — and why inference works
at all

— THE BIG IDEA & FORMAL STATEMENT

Central Limit Theorem — The Big Idea

If you take large enough samples ($n \geq 30$) from ANY population, sample means $\bar{x} \sim N(\mu, \sigma^2/n)$

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THREE POPULATIONS, ONE RESULT

CLT Visualised — Three Different Populations, One Result

Population shape does not matter — Right-skewed, Uniform, and Bimodal all become Normal sampling distributions

Right-Skewed (Income / Waits)

Uniform (Rolling Die)

Bimodal (Two Segments)

Sample Size per Draw (n):

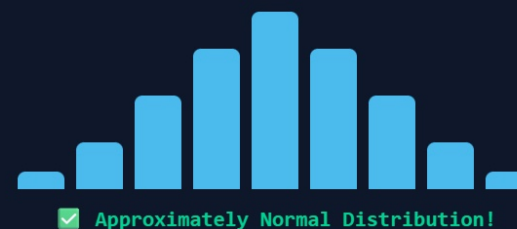


n = 36 (CLT Active!)

1. POPULATION DISTRIBUTION



2. SAMPLING DIST OF MEAN (X-BAR)



— QUALITY CONTROL APPLICATION

CLT Applied — Quality Control at a Factory

Scenario: Batteries with $\mu = 500$ hrs, $\sigma = 40$. Sample $n = 64$ has $\bar{x} = 488$ hrs. Is machine miscalibrated?

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— EXERCISE 03 · CENTRAL LIMIT THEOREM

Exercise — Central Limit Theorem

A logistics company delivers packages with $\mu = 3.2$ days, $\sigma = 1.5$ days (right-skewed). Sample $n = 36$ gives $\bar{x} = 3.6$ days.

Q1: Can we apply the CLT here? What does it tell us about the sampling distribution of $\bar{x}$ -bar?

Hint: Check whether $n \geq 30$. What will the sampling distribution look like even though population is skewed?

Reveal Solution ▾

Q2: Calculate the Standard Error (SE) of the sample mean.

Hint: $SE = \sigma / \sqrt{n}$. Plug in $\sigma = 1.5$ and $n = 36$.

Reveal Solution ▾

Q3: Calculate the z-score for $\bar{x} = 3.6$ days.

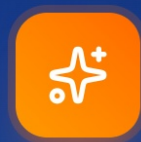
Hint: $z = (\bar{x} - \mu) / SE$. How many standard errors is 3.6 from 3.2?

Reveal Solution ▾

Q4: Does this sample provide strong evidence delivery times worsened? What probability are you computing?

Hint: $P(Z > \text{your } z\text{-score})$. $P(Z > 1.6)$ is approximately 0.055.

Reveal Solution ▾



05 • HOW UNCERTAIN ARE WE?

How Uncertain Are We?

Putting honest error bars around our estimates of the unknown

— MISCONCEPTION VS CORRECT MEANING

What Is a Confidence Interval?

Fixing the #1 most common misconception in statistics: Probability is about the procedure, not a single calculated interval.

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— $CI = \bar{X} \pm Z^* (\sigma / \sqrt{N})$

Confidence Interval — Formula and Full Worked Example

Customer Satisfaction Survey: $n = 100$, $\bar{x} = 7.8$, $s = 1.5$. 95% CI = (7.51, 8.09)

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— COMPARING 90%, 95%, AND 99% CI

Confidence Intervals in Product Analytics

Daily active usage: $n = 400$, $\bar{x} = 23$ min/day, $s = 8$ min. $SE = 0.4$. Trade-off between confidence & precision.

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— EXERCISE 04 · CONFIDENCE INTERVALS

Exercise — Confidence Intervals

A call centre monitors agent performance. Sample $n = 81$ calls gives average handling time $\bar{x} = 7.4$ min, $s = 2.7$ min.

Q1: Calculate the Standard Error (SE) for this sample.

Hint: $SE = s / \sqrt{n}$. Use $n = 81$ and $s = 2.7$.

Reveal Solution ▾

Q2: Build a 95% confidence interval for the true mean handling time.

Hint: Use $z^* = 1.96$. $CI = \bar{x} \pm z^* \times SE$.

Reveal Solution ▾

Q3: Target handling time is 7.0 min. Is there strong evidence agents exceed target?

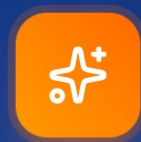
Hint: Does 7.0 fall inside or outside your interval? What does that tell you?

Reveal Solution ▾

Q4: The manager wants margin of error reduced to 0.20 min at 95% confidence. What sample size is needed?

Hint: $ME = z^* \times s / \sqrt{n}$. Rearrange for n : $n = (z^* \times s / ME)^2$.

Reveal Solution ▾



06 • REAL OR RANDOM?

Is This Result Real or Random?

The formal framework for making evidence-based decisions from data

— NULL (H_0) VS ALTERNATIVE (H_1)

The Logic of Hypothesis Testing — H_0 vs H_1

Innocent until proven guilty — We assume NO effect (H_0) until data proves otherwise (H_1)

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— WHAT IT IS VS WHAT IT IS NOT

The p-value — What It Really Means

p-value: Probability of observing a result at least as extreme as yours, IF H_0 were true.

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— TYPE I (ALPHA) VS TYPE II (BETA)

Type I and Type II Errors — Two Ways to Be Wrong

Understanding the cost of each mistake, controlling alpha, and power ($1 - \beta$)

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— DECISION TREE: Z-TEST VS T-TEST

z-test vs t-test — When to Use Which

Same core logic, different situations: Do you know population sigma?

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Same core logic, different situations: Do you know population sigma?

— THE 5-STEP FRAMEWORK

Hypothesis Testing — The 5-Step Framework

01. State Hypotheses -> 02. Set Alpha -> 03. Compute Test Stat -> 04. Find p-value -> 05. Business Decision

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— FULL HYPOTHESIS TEST WALKTHROUGH

Applied Example — Did the New Homepage Increase Revenue?

Baseline $\mu_0 = \$120$. Redesign sample $n = 36$, $\bar{x} = \$128$, $\sigma = \$24$. $z = 2.0$, $p = 0.0228 \rightarrow$ Reject H_0 !

Applied Example — Did the New Homepage Increase Revenue?

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— EXERCISE 05 · FULL HYPOTHESIS TEST

Exercise — Run a Full Hypothesis Test

Bank call center: $\mu_0 = 8$ min, $\sigma = 2$ min. After training, $n = 25$ calls has $\bar{x} = 7.2$ min. At $\alpha = 0.01$, did training work?

Step 01: State H_0 and H_1 explicitly.

Hint: Is this one-tailed or two-tailed? We expect a decrease in call time.

Reveal Solution ▾

Step 02 & 03: Choose test & calculate test statistic z .

Hint: σ is known = 2 min, $n = 25$. $z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$.

Reveal Solution ▾

Step 04: Find critical value z^* at $\alpha = 0.01$ (one-tailed left).

Hint: For $\alpha = 0.01$ left-tailed, $z^* = -2.326$. Is $z = -2.0$ more extreme?

Reveal Solution ▾

Step 05: State your business conclusion and recommendation.

Hint: Did training significantly reduce call time at 1% level?

Reveal Solution ▾

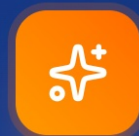
— FULL STEP-BY-STEP SOLUTION

Exercise Solution — Bank Call Time

Key lesson: A promising sample result (7.2 vs 8.0 min) can still be statistically insignificant at $\alpha = 0.01$ with $n = 25$.

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07 • WHICH VERSION WINS?

Which Version Actually Wins?

Turning hypothesis testing into a repeatable engine for digital product decisions

— CONTROL (A) VS TREATMENT (B)

Control Group vs Treatment Group

The two arms of every A/B test — Random assignment ensures causal comparison

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TWO-PROPORTION Z-TEST

The Two-Proportion Z-Test — Formula and Worked Example

Landing page CTA test: $n_A = 1000$ (120 conv = 12%), $n_B = 1000$ (150 conv = 15%). Pooled $p = 0.135$, $z = 1.96$, $p = 0.050$.

The Two-Proportion Z-Test — Formula and Worked Example

Landing page CTA test: $n_A = 1000$ (120 conv = 12%), $n_B = 1000$ (150 conv = 15%). Pooled $p = 0.135$, $z = 1.96$, $p = 0.050$.

CHECKOUT BUTTON COLOR TEST

Applied Example — Did the New Checkout Button Increase Conversions?

Control (Green): 5,000 visitors, 400 conv (8.0%). Treatment (Orange): 5,000 visitors, 460 conv (9.2%). $z = 2.14$, $p = 0.0162$ → Roll out!

INTERACTIVE A/B TEST INPUTS

Control (A)

Visitors (nA):

Conversions (xA):

Treatment (B)

Visitors (nB):

Conversions (xB):

RESULTS & Z-STATISTIC

p_A (Control) = 8.00%

p_B (Treatment) = 9.20%

Pooled p -hat = 8.60%

Pooled SE = 0.00561

z-statistic = 2.14



REJECT H_0 – Statistically Significant Lift!

Roll out the new variation to 100% of users!

— HOW MANY VISITORS DO YOU NEED?

Sample Size & Power — How Many Visitors Do You Need?

Under-powered tests are the #1 reason A/B results can't be trusted. Signup page lift 10% -> 12% needs ~7,666 visitors.

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— STATISTICAL PITFALLS

A/B Testing Pitfalls — Statistical Traps That Ruin Experiments

Running the test correctly is only half the job — most bad calls come from misreading valid results:



[PK] The Peeking Problem

Checking results repeatedly and stopping when $p < 0.05$ inflates false positives from 5% up to 30%.

Fix: Pre-register sample size or use sequential testing.



[MC] Multiple Comparisons

Testing 20 metrics at $\alpha = 0.05$ guarantees roughly 1 false positive by chance.

Fix: Apply Bonferroni correction (α / n) or control FDR.



[NV] Novelty Effect

Users react to a NEW feature simply because it is new. Lift fades in 1–2 weeks.

Fix: Run test long enough to see effect stabilize.



[SP] Simpson's Paradox

An overall winning result can reverse when split by segment (device, region, user type).

Fix: Always check segment-level results before rollout.



[UP] Underpowered Tests

Stopping early or small sample size means real effects go undetected.

Fix: Calculate required n for target MDE & 80% power before launching.



[SR] Sample Ratio Mismatch

If traffic isn't split as intended (e.g. 51/49 instead of 50/50), test is invalid due to randomization bug.

Fix: Run Chi-Square check on traffic split.

— WHY CHECKING EARLY INFLATES FALSE POSITIVES

The Peeking Problem — Why Checking Early Inflates False Positives

Even with ZERO true difference, 5 peeks raises false positive risk from 5% to 14%; 20 peeks raises it to 29%!

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— EXERCISE 06 · A/B TESTING

Exercise — A/B Testing

E-commerce product page test: Control A (2,500 visitors, 175 purchases), Treatment B (2,500 visitors, 210 purchases).

Q1: Calculate conversion rate for each group and pooled proportion $\hat{p}$.

Hint: $p_A = x_A/n_A$, $p_B = x_B/n_B$. $\hat{p} = (x_A + x_B)/(n_A + n_B)$.

Reveal Solution ▾

Q2: Calculate the z-statistic for this two-proportion test.

Hint: $SE = \sqrt{\hat{p}(1-\hat{p}) \times (1/n_A + 1/n_B)}$. $z = (p_B - p_A) / SE$.

Reveal Solution ▾

Q3: At $\alpha = 0.05$ (two-tailed, $z^* = 1.96$), is difference statistically significant?

Hint: Compare $|z|$ to 1.96. Should team roll out new layout?

Reveal Solution ▾

Q4: The team peeked after 500 visitors/group, saw $p = 0.04$, and stopped early. What is wrong?

Hint: Think about peeking problem and true false-positive rate.

Reveal Solution ▾

— SUMMARY OF CONCEPTS & RULES

A/B Testing — Quick Reference

Complete cheat sheet table of concepts, formulas, when to use, and key insights

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Complete cheat sheet table of concepts, formulas, when to use, and key insights

— COMPLETE INFERENCE STATISTICS CHEAT SHEET

Inferential Statistics — Complete Cheat Sheet

All 12 core concepts, formulas, decision rules, and key insights in one master reference table

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All 12 core concepts, formulas, decision rules, and key insights in one master reference table

YOU CAN NOW:

Congratulations! You have mastered Inferential Statistics, Probability Theory, Bayes' Theorem, CLT, Confidence Intervals, Hypothesis Testing, and A/B Testing.